Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Borges Farms

Physical address of dairy:

5148 W Grayson RD	Modesto	Stanislaus	95358
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1973

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

0017-0059-0006-0000 0017-0059-0007-0000 0017-0062-0003-0000

B. OPERATORS

Borges, Candido A.

Operator name: Borges, Candido A. Telephone no.: (209) 538-7810 Landline (209) 481-6363 Cellular

5148 W Grayson RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Borges, Candido A.

Legal owner name: Borges, Candido A. Telephone no.: (209) 538-7810 Landline (209) 481-6363 Cellular

5148 W Grayson RD	Modesto	CA	95358
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	135	150	100	100	150
Number under roof	750	0	0	0	0	0
Maximum number	750	135	150	100	100	150
Average number	750	135	150	100	100	150
Avg live weight (lbs)	1,400	1,450	850	550		

Predominant milk cow breed: Holstein

Average milk production: 60 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 23,583.99 *tons per reporting period*

Total nitrogen from manure: 301,164.12 *lbs per reporting period*

After ammonia losses (30% loss applied): 210,814.88 *lbs per reporting period*

Total phosphorus from manure: 48,637.81 *lbs per reporting period*

Total potassium from manure: 138,704.42 *lbs per reporting period*

Total salt from manure: 384,180.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 2,160,000 *gallons*

Total nitrogen generated: 7,570.58 *lbs*

Total phosphorus generated: 1,070.70 *lbs*

Total potassium generated: 14,330.03 *lbs*

Total salt generated: 87,241.97 *lbs*

	2,160,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	2,160,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Irrigation District	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/01/2023	Liquid commercial fertilizer *UN32*	17,370.00 *gal*			32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	46,384.85	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	46,384.85	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	15	15	1	both	X017-X059-X006-XXXX
10	39	39	1	manure	X017-X058-X025-XXXX
11	9	9	1	none	X017-X058-X033-XXXX
12	135	135	1	none	X017-X058-X055-XXXX
2	40	40	1	both	X017-X059-X006-XXXX
3	40	40	1	both	X017-X059-X007-XXXX
4	35	35	1	both	X017-X059-X007-XXXX
5	38	38	1	both	X017-X059-X007-XXXX
6	40	40	1	both	X017-X059-X007-XXXX
7	38	38	1	manure	X017-X062-X050-XXXX
8	37	37	1	manure	X017-X062-X051-XXXX
9	38	38	1	manure	X017-X058-X026-XXXX
Totals for areas that were used for application	360	360	10		
Totals for areas that were not used for application	144	144	2		
Land application area totals	504	504	12		

B. CROPS AND HARVESTS

1

Field name: 1

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	400.00 *ton*	Dry-weight		71.6	10,300.00	1,670.00	17,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.67	156.01	25.29	269.61	1,045.12

10

10

Field name: 10

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 39 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,070.00 *ton*	Dry-weight		70.8	9,780.00	1,640.00	17,100.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.44	156.70	26.28	273.99	640.90

11

Field name: 11

01/01/2005: Almond, in shell

Crop: Almond, in shell Acres planted: 9 Plant date: 01/01/2005

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	9.00 *ton*	Dry-weight		20.0	100,000.00	6,000.00	100,000.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.00	130.00	20.00	100.00	0.00
Total actual harvest content	1.00	160.00	9.60	160.00	160.00

12

Field name: 12

12

01/01/2005: Walnuts

Crop: Walnuts Acres planted: 135 Plant date: 01/01/2005

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	67.00 *ton*	Dry-weight		20.0	100,000.00	6,000.00	100,000.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	0.50	50.00	10.00	50.00	0.00
Total actual harvest content	0.50	79.41	4.76	79.41	79.41

2

Field name: 2

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,100.00 *ton*	Dry-weight		71.4	10,900.00	1,150.00	14,300.00		5.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.50	171.46	18.09	224.94	849.42

3

Field name: 3

3

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,100.00 *ton*	Dry-weight		72.0	11,800.00	1,640.00	17,700.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.50	181.72	25.26	272.58	862.40

4

Field name: 4

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	950.00 *ton*	Dry-weight		71.8	11,400.00	1,490.00	17,300.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.14	174.52	22.81	264.84	887.90

5

Field name: 5

5

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,040.00 *ton*	Dry-weight		70.2	11,500.00	1,490.00	16,500.00		5.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.37	187.58	24.30	269.14	897.14

6

Field name: 6

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 40 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,090.00 *ton*	Dry-weight		70.2	12,300.00	1,420.00	13,400.00		5.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.25	199.76	23.06	217.63	938.73

7

Field name: 7

7

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,030.00 *ton*	Dry-weight		72.6	11,500.00	1,320.00	15,300.00		4.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.11	170.82	19.61	227.26	727.83

8

Field name: 8

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,000.00 *ton*	Dry-weight		69.6	11,100.00	1,270.00	15,900.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.03	182.40	20.87	261.28	838.05

9

Field name: 9

9

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/18/2023	1,040.00 *ton*	Dry-weight		71.0	11,300.00	2,320.00	13,900.00		4.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.37	179.37	36.83	220.64	650.82

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 06/12/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	48.56	15.89	104.13	0.00	25.00 *ton*
Application event totals		48.56	15.89	104.13	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.84	0.00	0.00	0.00	1,817,766.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

1 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	33.65	4.76	63.69	387.74	144,000.00 *gal*
Application event totals		33.65	4.76	63.69	387.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.53	0.00	0.00	0.00	1,615,792.00 *gal*
Application event totals		2.53	0.00	0.00	0.00	

10 - 06/12/2023: Corn, silage

10 - 06/12/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	104.58	34.22	224.27	0.00	140.00 *ton*
Application event totals		104.58	34.22	224.27	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.91	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.91	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

10 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.43	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.43	0.00	0.00	0.00	

11 - 01/01/2005: Almond, in shell

Field name: 11

Crop: Almond, in shell

Plant date: 01/01/2005

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

11 - 01/01/2005: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	150.00	0.00	0.00	0.00	
Application event totals		150.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

11 - 01/01/2005: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	807,896.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

12 - 01/01/2005: Walnuts

Field name: 12

Crop: Walnuts

Plant date: 01/01/2005

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	120.00	0.00	0.00	0.00	
Application event totals		120.00	0.00	0.00	0.00	

05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

12 - 01/01/2005: Walnuts

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.10	0.00	0.00	0.00	12,118,441.00 *gal*
Application event totals		2.10	0.00	0.00	0.00	

2 - 06/12/2023: Corn, silage

2 - 06/12/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	72.83	23.83	156.19	0.00	100.00 *ton*
Application event totals		72.83	23.83	156.19	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.84	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,380.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	37.85	5.35	71.65	436.21	432,000.00 *gal*
Application event totals		37.85	5.35	71.65	436.21	

2 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

3 - 06/12/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	72.83	23.83	156.19	0.00	100.00 *ton*
Application event totals		72.83	23.83	156.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.84	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	37.85	5.35	71.65	436.21	432,000.00 *gal*
Application event totals		37.85	5.35	71.65	436.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

3 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

4 - 06/12/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	83.24	27.23	178.50	0.00	100.00 *ton*
Application event totals		83.24	27.23	178.50	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.71	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.71	0.00	0.00	0.00	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	36.05	5.10	68.24	415.44	360,000.00 *gal*
Application event totals		36.05	5.10	68.24	415.44	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

4 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.44	0.00	0.00	0.00	3,635,532.00 *gal*
Application event totals		2.44	0.00	0.00	0.00	

5 - 06/12/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	76.67	25.08	164.41	0.00	100.00 *ton*
Application event totals		76.67	25.08	164.41	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.74	0.00	0.00	0.00	4,443,428.00 *gal*
Application event totals		2.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	33.20	4.70	62.85	382.64	360,000.00 *gal*
Application event totals		33.20	4.70	62.85	382.64	

5 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

6 - 06/12/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	72.83	23.83	156.19	0.00	100.00 *ton*
Application event totals		72.83	23.83	156.19	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.84	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Wastewater	Process wastewater	37.85	5.35	71.65	436.21	432,000.00 *gal*
Application event totals		37.85	5.35	71.65	436.21	

6 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.37	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.37	0.00	0.00	0.00	

7 - 06/12/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

7 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	107.33	35.12	230.18	0.00	140.00 *ton*
Application event totals		107.33	35.12	230.18	0.00	

06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.99	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.99	0.00	0.00	0.00	

07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

7 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

8 - 06/12/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	110.23	36.06	236.40	0.00	140.00 *ton*
Application event totals		110.23	36.06	236.40	0.00	

8 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	3.07	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		3.07	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

8 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.56	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.56	0.00	0.00	0.00	

9 - 06/12/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Manure	Corral solids	107.33	35.12	230.18	0.00	140.00 *ton*
Application event totals		107.33	35.12	230.18	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.99	0.00	0.00	0.00	4,847,376.00 *gal*
Application event totals		2.99	0.00	0.00	0.00	

9 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Liquid commercial fertilizer	80.00	0.00	0.00	0.00	
Application event totals		80.00	0.00	0.00	0.00	

07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

9 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

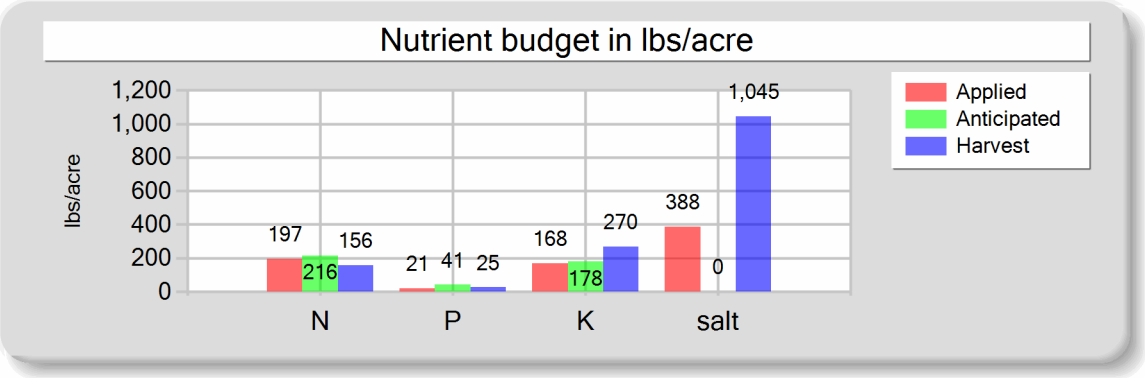
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation District	Surface water	2.49	0.00	0.00	0.00	4,039,480.00 *gal*
Application event totals		2.49	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 06/12/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	48.56	15.89	104.13	0.00
Process wastewater	33.65	4.76	63.69	387.74
Fresh water	20.52	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	196.73	20.64	167.82	387.74
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	156.01	25.29	269.61	1,045.12
Nutrient balance	40.72	-4.65	-101.79	-657.38
Applied to removed ratio	1.26	0.82	0.62	0.37

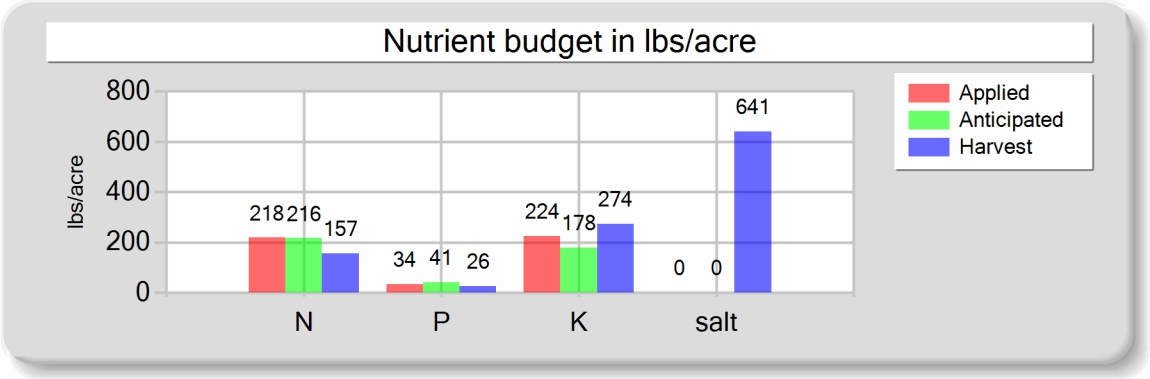
Fresh water applied	
13,128,310.00	*gallons*
483.47	*acre-inches*
32.23	*inches/acre*

Process wastewater applied	
144,000.00	*gallons*
5.30	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/12/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	104.58	34.22	224.27	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	19.92	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	218.50	34.22	224.27	0.00
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	156.70	26.28	273.99	640.90
Nutrient balance	61.80	7.94	-49.71	-640.90
Applied to removed ratio	1.39	1.30	0.82	0.00

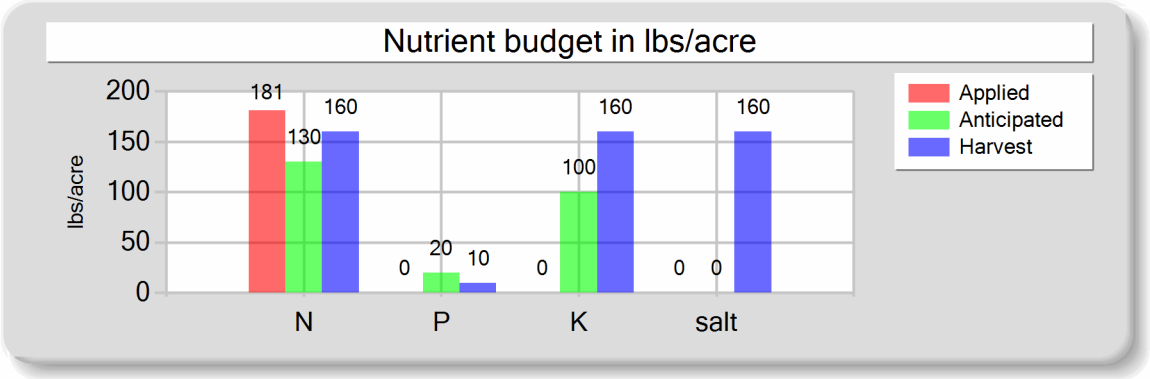
Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
31.28	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 01/01/2005: Almond, in shell

Field name: 11 Crop: Almond, in shell Plant date: 01/01/2005



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	150.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.84	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	180.84	0.00	0.00	0.00
Anticipated crop nutrient removal	130.00	20.00	100.00	0.00
Actual crop nutrient removal	160.00	9.60	160.00	160.00
Nutrient balance	20.84	-9.60	-160.00	-160.00
Applied to removed ratio	1.13	0.00	0.00	0.00

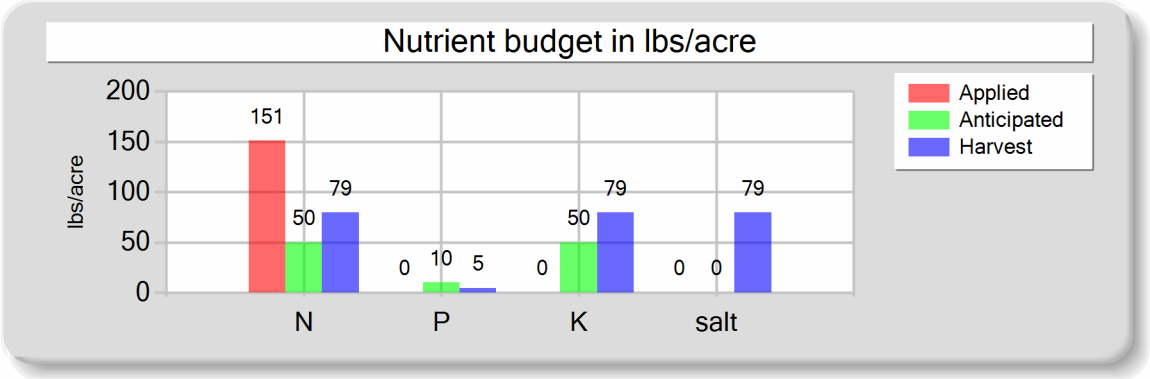
Fresh water applied	
6,463,168.00	*gallons*
238.02	*acre-inches*
26.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

12 - 01/01/2005: Walnuts

Field name: 12 Crop: Walnuts Plant date: 01/01/2005



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	120.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.84	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	150.84	0.00	0.00	0.00
Anticipated crop nutrient removal	50.00	10.00	50.00	0.00
Actual crop nutrient removal	79.41	4.76	79.41	79.41
Nutrient balance	71.43	-4.76	-79.41	-79.41
Applied to removed ratio	1.90	0.00	0.00	0.00

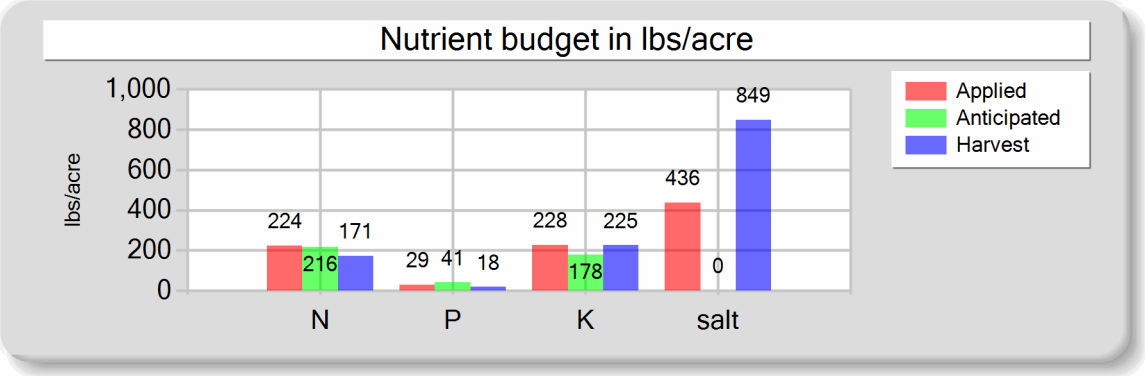
Fresh water applied	
96,947,528.00	*gallons*
3,570.25	*acre-inches*
26.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/12/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	72.83	23.83	156.19	0.00
Process wastewater	37.85	5.35	71.65	436.21
Fresh water	19.42	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	224.10	29.18	227.84	436.21
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	171.46	18.09	224.94	849.42
Nutrient balance	52.65	11.09	2.90	-413.21
Applied to removed ratio	1.31	1.61	1.01	0.51

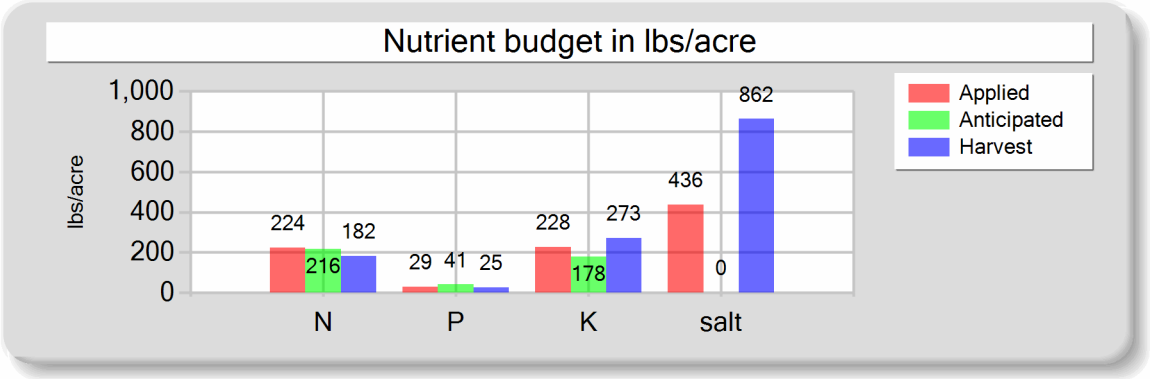
Fresh water applied	
33,123,636.00	*gallons*
1,219.83	*acre-inches*
30.50	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/12/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	72.83	23.83	156.19	0.00
Process wastewater	37.85	5.35	71.65	436.21
Fresh water	19.42	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	224.10	29.18	227.84	436.21
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	181.72	25.26	272.58	862.40
Nutrient balance	42.38	3.93	-44.74	-426.19
Applied to removed ratio	1.23	1.16	0.84	0.51

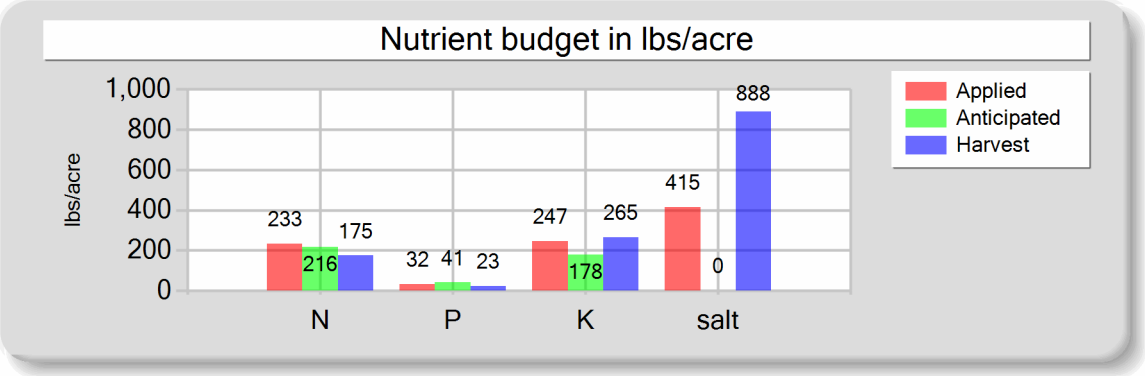
Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
30.50	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/12/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	83.24	27.23	178.50	0.00
Process wastewater	36.05	5.10	68.24	415.44
Fresh water	19.76	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	233.04	32.33	246.74	415.44
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	174.52	22.81	264.84	887.90
Nutrient balance	58.53	9.52	-18.10	-472.46
Applied to removed ratio	1.34	1.42	0.93	0.47

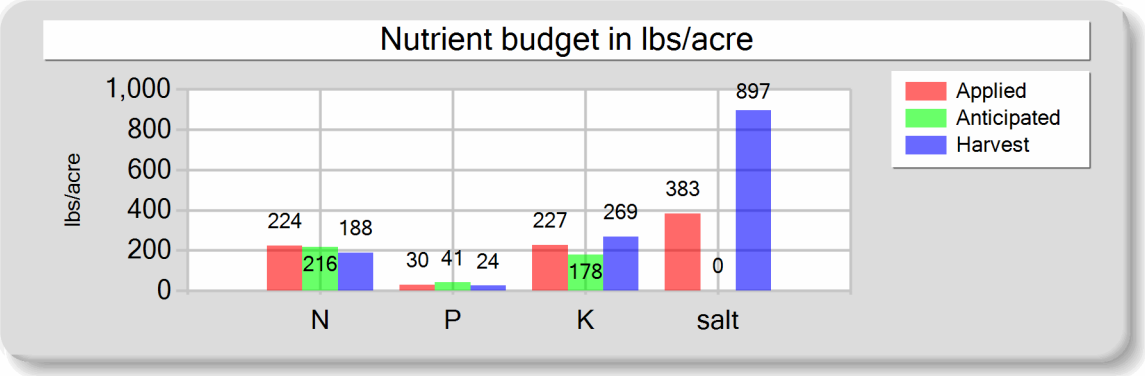
Fresh water applied	
29,488,204.00	*gallons*
1,085.95	*acre-inches*
31.03	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.38	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/12/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	76.67	25.08	164.41	0.00
Process wastewater	33.20	4.70	62.85	382.64
Fresh water	20.19	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	224.06	29.78	227.26	382.64
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	187.58	24.30	269.14	897.14
Nutrient balance	36.48	5.47	-41.88	-514.50
Applied to removed ratio	1.19	1.23	0.84	0.43

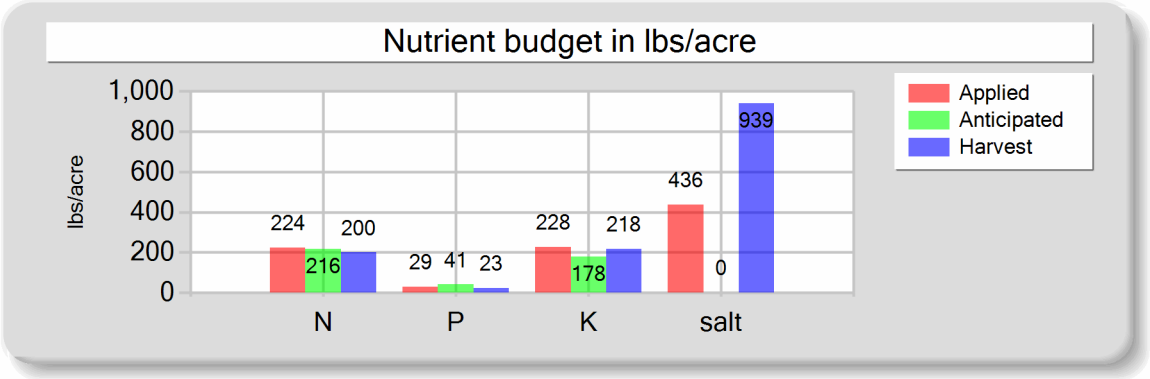
Fresh water applied	
32,719,788.00	*gallons*
1,204.96	*acre-inches*
31.71	*inches/acre*

Process wastewater applied	
360,000.00	*gallons*
13.26	*acre-inches*
0.35	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 06/12/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	72.83	23.83	156.19	0.00
Process wastewater	37.85	5.35	71.65	436.21
Fresh water	19.42	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	224.10	29.18	227.84	436.21
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	199.76	23.06	217.63	938.73
Nutrient balance	24.34	6.12	10.21	-502.52
Applied to removed ratio	1.12	1.27	1.05	0.46

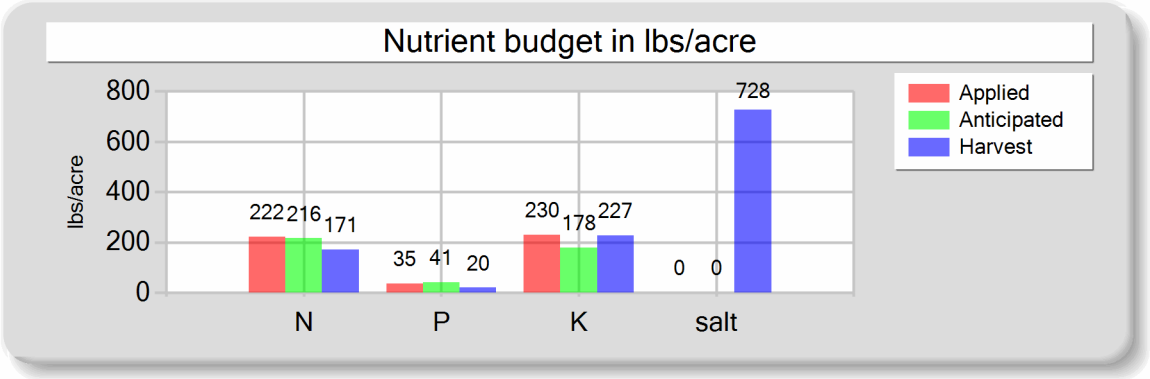
Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
30.50	*inches/acre*

Process wastewater applied	
432,000.00	*gallons*
15.91	*acre-inches*
0.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 06/12/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	107.33	35.12	230.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.44	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	221.77	35.12	230.18	0.00
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	170.82	19.61	227.26	727.83
Nutrient balance	50.96	15.51	2.91	-727.83
Applied to removed ratio	1.30	1.79	1.01	0.00

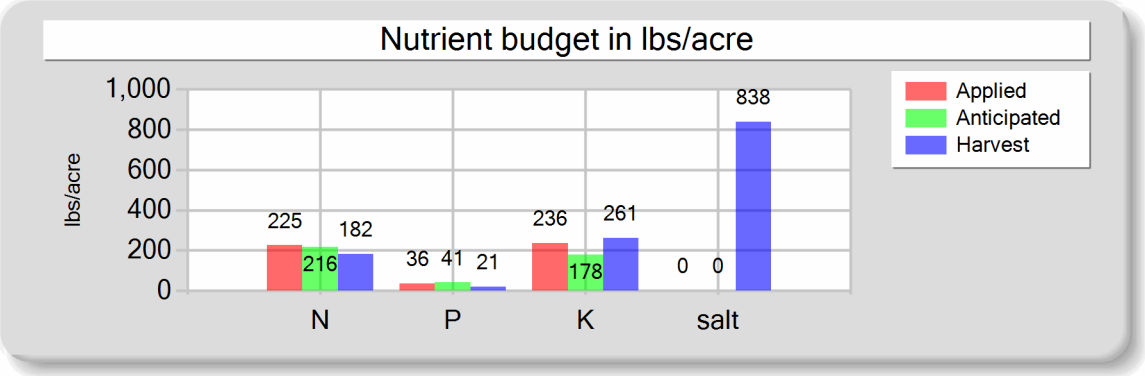
Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
32.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/12/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	110.23	36.06	236.40	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.99	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	225.23	36.06	236.40	0.00
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	182.40	20.87	261.28	838.05
Nutrient balance	42.83	15.20	-24.88	-838.05
Applied to removed ratio	1.23	1.73	0.90	0.00

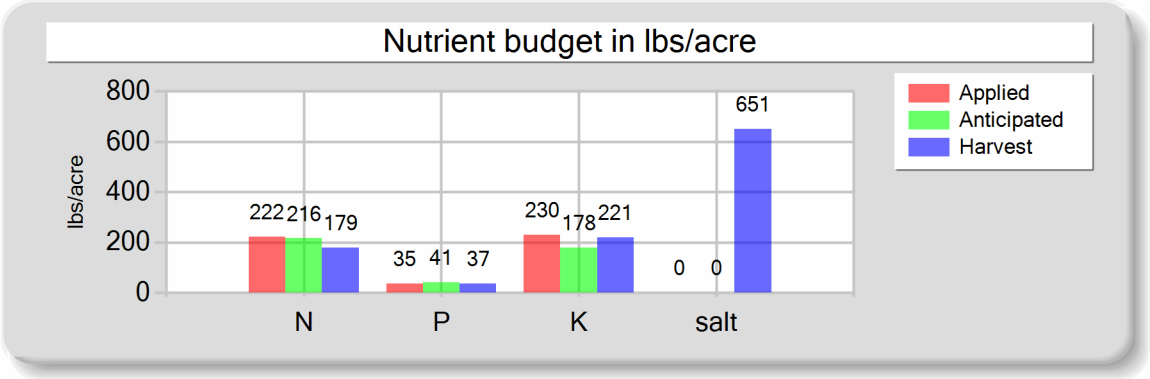
Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
32.97	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 06/12/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	80.00	0.00	0.00	0.00
Dry manure	107.33	35.12	230.18	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.44	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	221.77	35.12	230.18	0.00
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	179.37	36.83	220.64	650.82
Nutrient balance	42.40	-1.71	9.53	-650.82
Applied to removed ratio	1.24	0.95	1.04	0.00

Fresh water applied	
33,123,736.00	*gallons*
1,219.83	*acre-inches*
32.10	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 01/26/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 15.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	5,660.00	37,100.00							
DL	100.00	10.00	100.00							

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 07/27/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 18.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,100.00	5,600.00	33,000.00							
DL	100.00	10.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.
Sample date: 01/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	249.00	0.00	0.00	88.30	899.00								6,580.00	4,400
DL	1.00	1.00	1.00	1.00	1.00	1.00								1.00	1

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	420.00	249.00	0.00	0.00	59.40	795.00								6,760.00	4,840
DL	1.00	1.00	1.00	1.00	0.10	1.00								1.00	1

C. FRESH WATER ANALYSES

Irrigation District

Denele Analytical, Inc.

Sample description: Denele Analytical, Inc.

Sample date: 05/04/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.81	0.00	0.00								0.00	0
DL	0.01	0.50	0.14								1.00	1

D. SOIL ANALYSES

1

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00				
DL			1.00				

10

10

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00				
DL			1.00				

11

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			60.00				
DL			1.00				

12

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			63.00				
DL			1.00				

2

2

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			1.00				

3

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00				
DL			1.00				

4

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			82.00				
DL			1.00				

5

5

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00				
DL			1.00				

6

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			58.00				
DL			1.00				

7

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00				
DL			1.00				

8

8

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			21.00				
DL			1.00				

9

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 07/27/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			45.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

1 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	1,670.00	17,800.00		6.90
DL	100.00	10.00	100.00		0.10

10 - 06/12/2023: Corn, silage

10 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	9,780.00	1,640.00	17,100.00		4.00
DL	10.00	10.00	100.00		0.10

11 - 01/01/2005: Almond, in shell

Estimate - No lab analysis

Sample and source description: Estimate - No lab analysis

Sample date: 10/15/2023 Source of analysis: Other/estimated Method of reporting: Dry-weight

Moisture: 20.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	100,000.00	6,000.00	100,000.00		10.00
DL	100.00	10.00	100.00		0.10

12 - 01/01/2005: Walnuts

Estimate - No lab analysis

Sample and source description: Estimate - No lab analysis

Sample date: 10/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 20.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	100,000.00	6,000.00	100,000.00		10.00
DL	100.00	100.00	100.00		0.10

2 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,900.00	1,150.00	14,300.00		5.40
DL	100.00	10.00	100.00		0.10

3 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	1,640.00	17,700.00		5.60
DL	100.00	10.00	100.00		0.10

4 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	1,490.00	17,300.00		5.80
DL	100.00	10.00	100.00		0.10

5 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,490.00	16,500.00		5.50
DL	100.00	10.00	100.00		0.10

6 - 06/12/2023: Corn, silage

Denele Analytical, inc.

Sample and source description: Denele Analytical, inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,300.00	1,420.00	13,400.00		5.78
DL	100.00	10.00	100.00		0.10

7 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	1,320.00	15,300.00		4.90
DL	100.00	10.00	100.00		0.10

8 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	1,270.00	15,900.00		5.10
DL	100.00	10.00	100.00		0.10

9 - 06/12/2023: Corn, silage

Denele Analytical, Inc.

Sample and source description: Denele Analytical, Inc.

Sample date: 10/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	2,320.00	13,900.00		4.10
DL	100.00	10.00	100.00		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

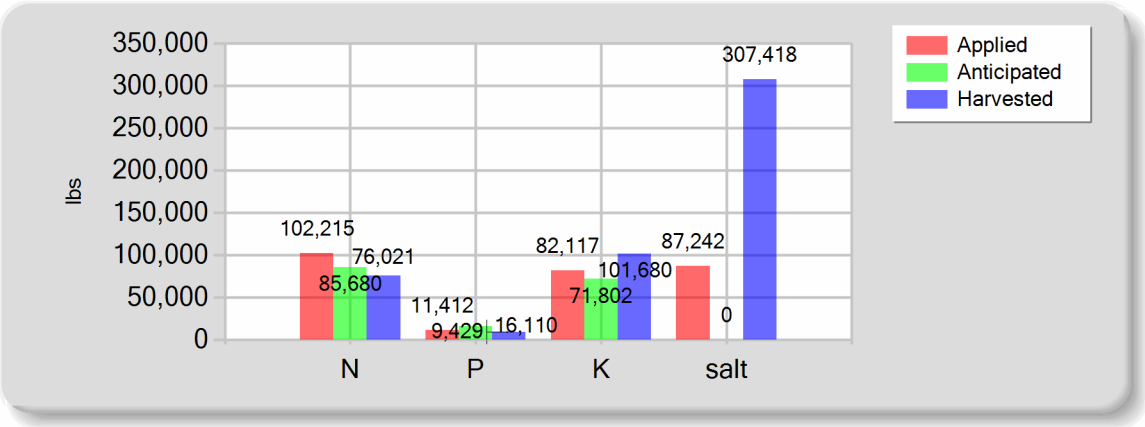
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

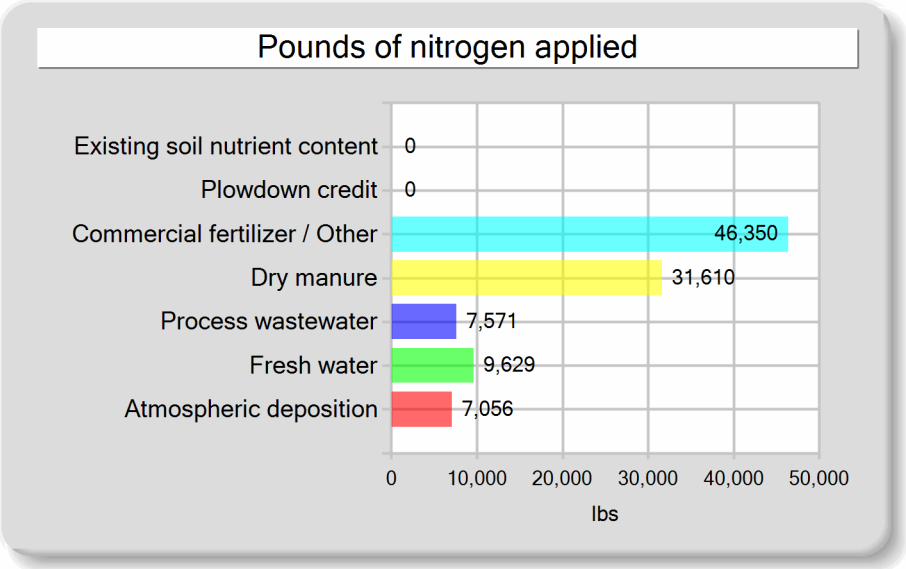
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

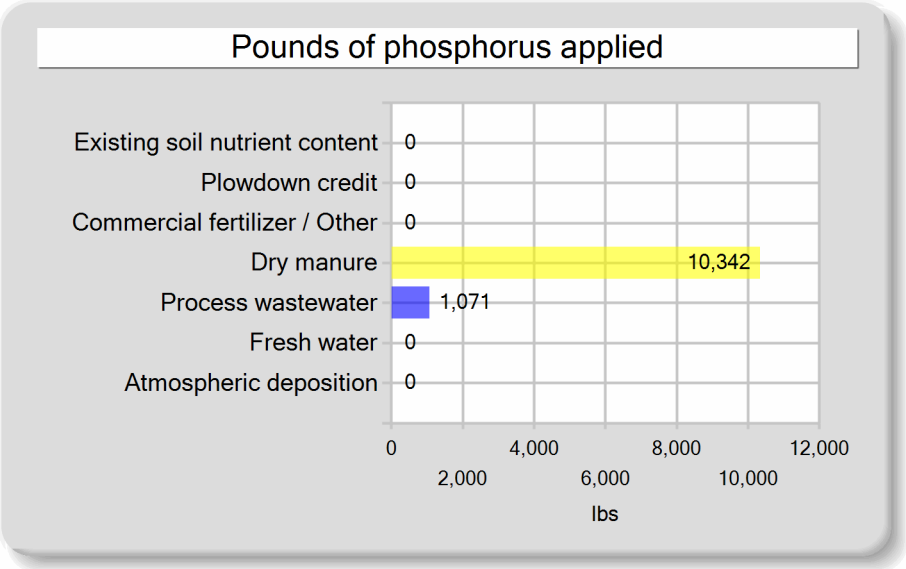
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	46,350.00	0.00	0.00	0.00
Dry manure	31,609.52	10,341.61	67,786.89	0.00
Process wastewater	7,570.58	1,070.70	14,330.03	87,241.97
Fresh water	9,628.65	0.00	0.00	0.00
Atmospheric deposition	7,056.00	0.00	0.00	0.00
Total nutrients applied	102,214.76	11,412.31	82,116.93	87,241.97
Anticipated crop nutrient removal	85,680.00	16,110.00	71,802.00	0.00
Actual crop nutrient removal	76,021.44	9,428.68	101,679.86	307,418.35
Nutrient balance	26,193.32	1,983.62	-19,562.93	-220,176.38
Applied to removed ratio	1.34	1.21	0.81	0.28

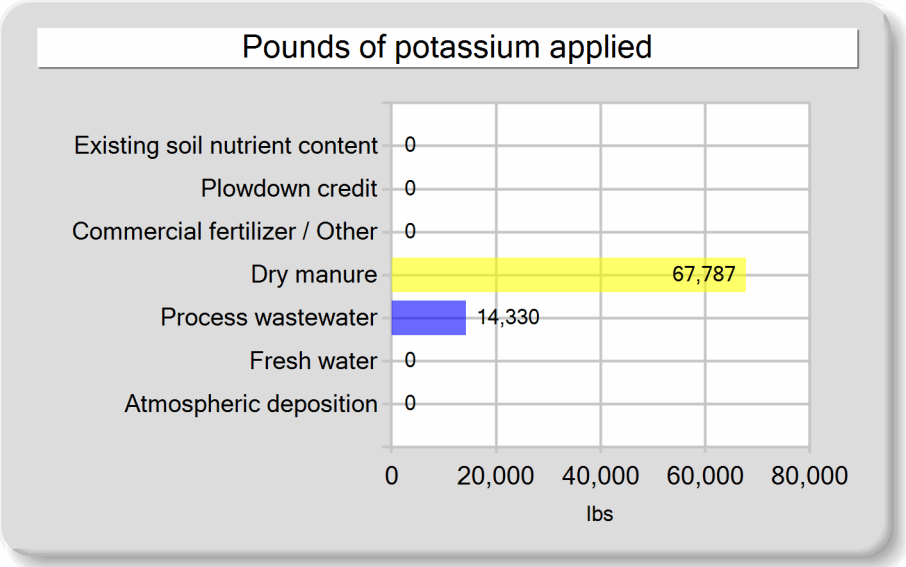
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

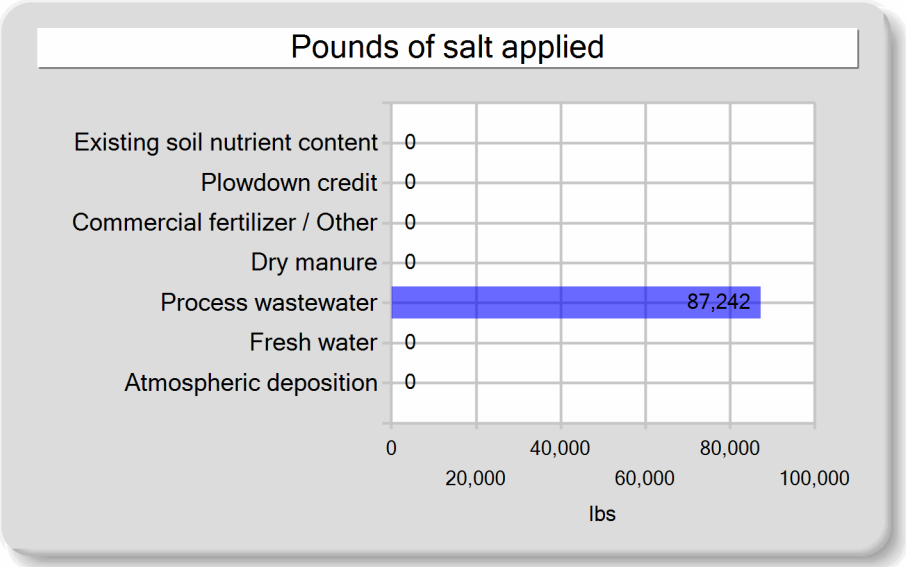


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Almonds and Walnuts were not sampled for lab analysis. No manure or manure water were applied to those parcels, 11 & 12.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Candido A. Borges	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Candido A. Borges	
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Candido A. Borges	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-18-24	
DATE	DATE

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Borges Farms
5148 W. Grayson Rd.
Modesto, CA 95358

Grower

Consultant:
Bruce Goldsmith
PO BOX 1056
Galt, CA 95632

PURCHASE ORDER:
RECEIVED DATE: 5/4/23 3:17 PM
SUBMITTED BY: Bruce
ANALYZED DATE: 5/8/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 18.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Irrigation District Water	T3124034A	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	< 10.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	2.81	mg/L
		Soluble Salts (SALT-SOL)	Calculation	55.0	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	101					
Electrical Conductivity (EC)	100			4.19 (T3123068J)	4.18 (Dup)	0.239
Nitrate Nitrogen (NO3-N)	102	100 (T3123010A)	106 (T3123010A)			5.55
Total Dissolved Solids (TDS)	103			0.000 (T3124034A)	0.000 (Dup)	
Total Nitrogen (TN)	96.3					

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: BORGES FARMS DAIRY
Mailing address: 5148 W. GRAYSON ROAD
(not facility address) MODESTO, CA 95358
Phone#: 209-481-6363

Authorized Copy Released To: BRUCE GOLDSMITH

Sampled By: BRUCE GOLDSMITH
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	5-4-23	2:45	IRRIGATION DISTRICT WATER		18.0						H7
2											
3											
4											
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
1st	[signature]	DAIRY COMPLIANCE GROUP				
2nd	[signature]	DENELE ANALYTICAL, INC.	5-4-23 3:17			T3124034

(LABORATORY USE ONLY)

Required to be submitted to any agency? (YES / NO)